

**Niro**

PATIENT-OWNED HEALTH RECORDS · BANGLADESH

IUT Techathon Nationals & Rover Summit 2026

Project Showcasing · Round 1 Abstract

Niro — Your Health, in Your Own Hands

আপনার স্বাস্থ্য, আপন হাতে

Team Cortex Crew

Senior Category

AI/ML — Digital Health

Fully working product

■ The problem — every household knows it

In Bangladesh, a family's entire medical history lives in a polythene bag: handwritten prescriptions no one can read, lab reports from different labs in different formats. Nobody — not the patient, not the doctor — can answer the simplest question about a life: *"Is my blood sugar getting better or worse?"* Every report is read once and forgotten. Doctors repeat tests that were already paid for, slow deteriorations creep by unnoticed, and short visits are spent reconstructing history from paper.

■ Our solution

Niro turns that bag of paper into a living, self-organizing health record — in Bangla. Two core innovations:

1

Auto-categorized records

স্বাস্থ্য রেকর্ড

Photograph any medical paper. A multimodal vision-language model reads it, explains it in plain Bangla, classifies it — prescription, CBC, blood-sugar report, lipid profile — extracts its date, and files it on the right shelf automatically. No typing, no prompting skill: the filing cabinet builds itself.

2

Health Trends

স্বাস্থ্য ট্রেন্ড

Every lab value is extracted into one of 28 canonical health metrics with its reference range, then charted across months. Abnormal values are flagged, and each metric carries a directional insight — *"fasting glucose rising over six months."* Scattered one-time readings become a visible trajectory that patients act on and doctors trust. No paper file — and no chatbot — can do this.

■ Around this core

The AI reads each new document against the patient's full history, flagging risks like a new drug clashing with an existing one; one tap requests a BMDC-licensed doctor's verification for 200–800 taka — the AI pre-writes the case summary, so review takes five minutes, not thirty; and in offline chambers a QR scan shares the record for the visit only, with time-bound, auto-revoking consent.

■ Safe by design

Niro never diagnoses and never prescribes; it extracts, explains, and flags. Low-confidence outputs automatically recommend human review, a banned-phrase linter blocks dosing language, every AI call and record access lands in an append-only audit log, and patients can delete all their data — aligned with Bangladesh's Data Protection Act 2023.

■ Uniqueness

Praava, Daktarbhai, Tonic, and Arogga store files, list doctors, or sell telemedicine calls; chatbots answer one question and forget you. None turns paper into a categorized, trend-tracking, patient-owned record in Bangla. Niro does.

■ Feasibility and cost

Fully built and working end-to-end (FastAPI, Next.js PWA, PostgreSQL) — not a concept. It runs in any phone browser, no app store needed; hosting fits free cloud tiers, and per-analysis AI cost is a few taka — near-zero operating cost.

Impact. For the first time, 170 million people can see their whole health story — organized, charted, and explained in their own language — and every doctor gets that story in thirty seconds. We are ready to demonstrate the complete live system at the final round.